`bluertopo`: Reproducible access to NOAA BlueTopo bathymetry in R

13 July 2026

Summary

Bathymetry describes the elevation of the submerged landscape: seabed, lakebed, and channel relief. NOAA BlueTopo is a public, tiled bathymetric product from the National Bathymetric Source program for U.S. waters and navigational lakes (U.S. Office of Coast Survey n.d.a, n.d.d). Its source data, quality context, and availability change over time, which makes a map alone an incomplete research record.

`bluertopo` is an R package for discovering current BlueTopo tiles that intersect a user-defined area, downloading original GeoTIFF and Raster Attribute Table (RAT) sidecar assets, checking their SHA-256 values, and opening selected bands as file-backed `terra` objects. It also retains catalog, query, selection, coverage, and download information when users request a detailed result. The package is intended for researchers and analysts who need to move from a geographic question to inspectable BlueTopo inputs in an R workflow, while keeping acquisition and resampling decisions explicit. It is not navigation software and does not perform vertical-datum transformations.

Statement of need

BlueTopo is distributed as evolving tiles rather than a single immutable analysis file. A reproducible workflow therefore needs more than a raster reader: it must identify the catalog version queried, preserve the specific source assets selected, distinguish a native cell size from an output-grid choice, and retain enough information to examine coverage and verification later. NOAA documents multiple access routes, including an AWS bucket, nowCOAST services, and download utilities (U.S. Office of Coast Survey n.d.b). Those access methods are valuable, but an R analysis still needs a transparent bridge from an AOI to local, verified assets and downstream spatial objects.

A later query may encounter a revised catalog, changed tile delivery date, or a different resolution mix. For that reason, the reproducible unit is not merely an AOI coordinate pair: it is the AOI together with the selection policy, catalog information, source URLs, verification result, and any requested output grid. Preserving these operational facts gives collaborators a practical basis for checking which data entered a result without pretending that an evolving service is a fixed local dataset.

`bluertopo` serves R users conducting coastal and marine spatial analysis, habitat or geomorphic context mapping, planning studies, and model input preparation. It does not interpret the bathymetry as a scientific result, estimate terrain derivatives, or convert vertical datums. NOAA specifies that BlueTopo can contain preliminary information, sources of differing quality, and interpolated areas; it also states that the product is not for measurement or navigation (U.S. Office of Coast Survey n.d.c). These boundaries are central to the package rather than cautions added after import.

State of the field

R spatial software provides mature foundations for geometry and raster work. The `sf` framework standardizes vector features and interoperates with external spatial systems (Pebesma 2018), while the modern R spatial ecosystem has continued to evolve around data representations and interfaces (Bivand 2021). Bathymetry-focused R software such as `marmap` supports importing, plotting, and analysing bathymetric and topographic data (Pante and Simon-Bouhet 2013). Gridded products such as SRTM15+ provide useful global bathymetric and topographic context (Tozer et al. 2019).

These tools address analysis or broad data access, not the specific problem of resolving an AOI against the current BlueTopo tile scheme and maintaining a verifiable local record of the selected NOAA assets. Conversely, NOAA supplies the data and several official access mechanisms (U.S. Office of Coast Survey n.d.b), not an R package that defines an analysis-facing provenance contract. Building a focused package instead of adding a few download calls to a general spatial package permits BlueTopo-specific choices—catalog schema checks, native-resolution policy, asset-sidecar handling, and source verification—to be tested and documented at one boundary. `bluertopo` delegates spatial operations to `terra`; it does not reimplement a general GIS stack.

Software design

The package separates discovery, acquisition, and spatial extraction. First, `bluertopo_tiles()` intersects the AOI with NOAA's current tile-scheme catalog, records native resolution and geometric coverage diagnostics, and applies an explicit selection policy. Figure 1 shows this step for a compact Key West AOI retrieved on 13 July 2026: one 4 m source tile was selected. Keeping this selection separate from downloading makes it possible to inspect what will be used before transferring data.

Coverage diagnostics describe the geometric intersection of the selected tile footprints with published coverage, rather than claiming that every cell has a particular scientific quality. This distinction matters when a resolution preference leaves an AOI only partially covered: the user can accept the documented coverage, request an error, or add lower-priority source resolutions through an explicit fill policy.

Second, the default download path retains the original GeoTIFF and available RAT sidecar, verifies against catalog-provided SHA-256 values, and writes CSV and JSON manifests. This design favors a durable source record over silently turning a remote tile into an untraceable local raster. The manifest also captures catalog identity, retrieval time, source URLs, coverage, and query information. The Key West run downloaded a verified 8.9 MB GeoTIFF and its RAT sidecar into the manuscript-local cache; its metadata is retained with the reproduction materials.

Third, native source selection is deliberately distinct from resampling. The package groups compatible source grids and otherwise returns a collection by default. A single output requires the user to supply both a projected CRS and an output resolution, at which point the package records that resampling occurred. This avoids treating distinct UTM zones, origins, or resolutions as though they were automatically interchangeable. Figure 2 is a display of the selected source elevation layer, not a new bathymetric inference. NOAA describes BlueTopo as a three-layer GeoTIFF product with elevation, uncertainty, and contributor information (U.S. Office of Coast Survey n.d.c).

Finally, a user can request the uncertainty band alongside elevation without discarding the source-file context. Figure 3 displays the associated vertical-uncertainty layer. NOAA notes that uncertainty and RAT metadata describe aspects of source quality and interpolation; the package does not convert those values into a navigation or fitness-for-purpose claim (U.S. Office of Coast Survey n.d.c, n.d.d). Cache deletion is also deliberately conservative: only a configured, package-marked cache can be cleared.

Research impact statement

The current package provides a documented, test-backed, reproducible route from AOI to verified BlueTopo assets and `terra` outputs. The audit at the manuscript commit found automated tests for catalog normalization, resolution and coverage policy, checksum-verified GeoTIFF/RAT retrieval, cache safeguards, and mixed grid handling; CI is configured across R versions and operating systems. These are credible near-term community-readiness signals for a BlueTopo-specific R workflow.

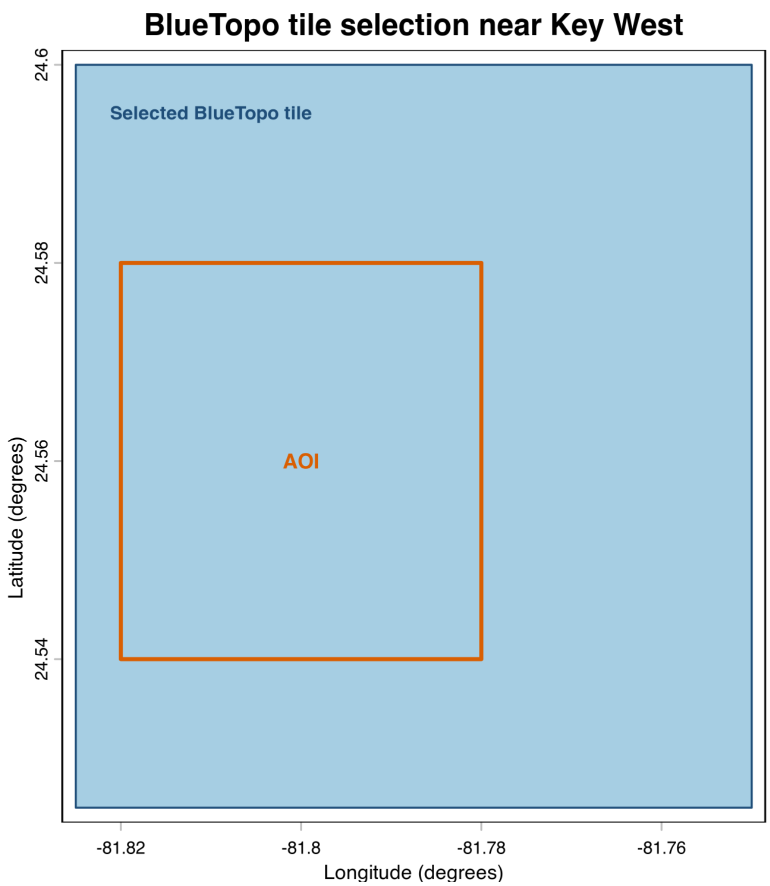

Figure 1: One 4 m BlueTopo tile footprint selected for the Key West AOI (-81.82, 24.54, -81.78, 24.58; WGS 84) on 13 July 2026. The blue polygon is the selected NOAA tile and the orange rectangle is the requested analysis area.

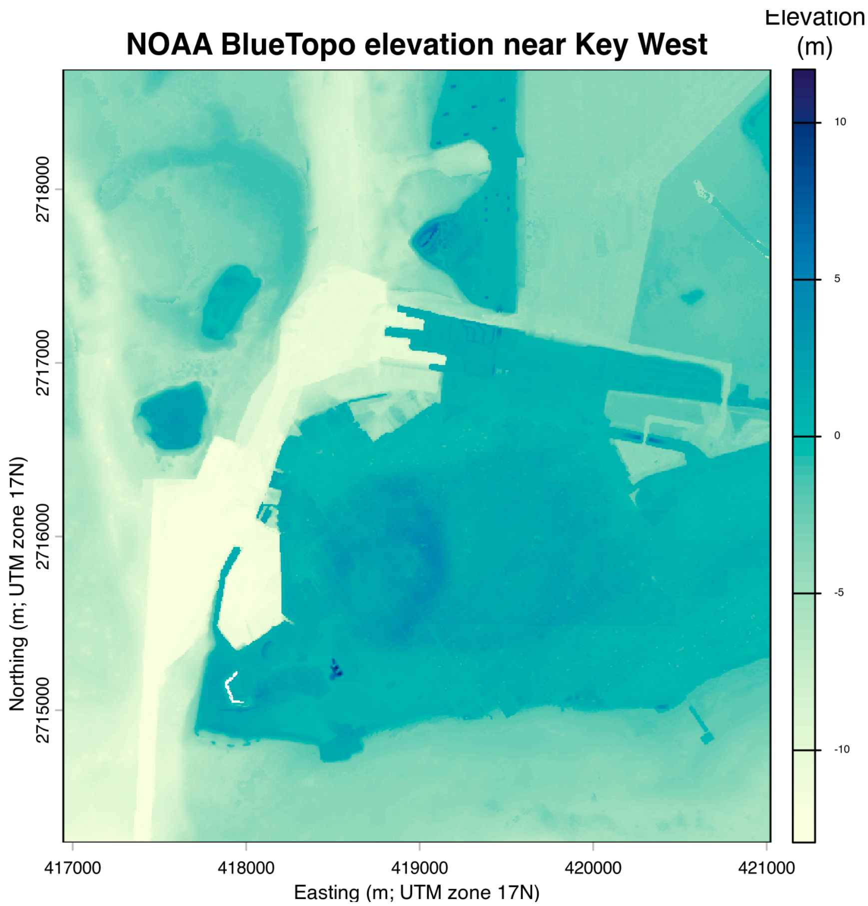

Figure 2: BlueTopo elevation for the Key West AOI, shown in the source tile's UTM zone 17N coordinates. The color scale is elevation in metres; the orange outline is the requested AOI. This is a software-behavior demonstration, not a scientific interpretation.

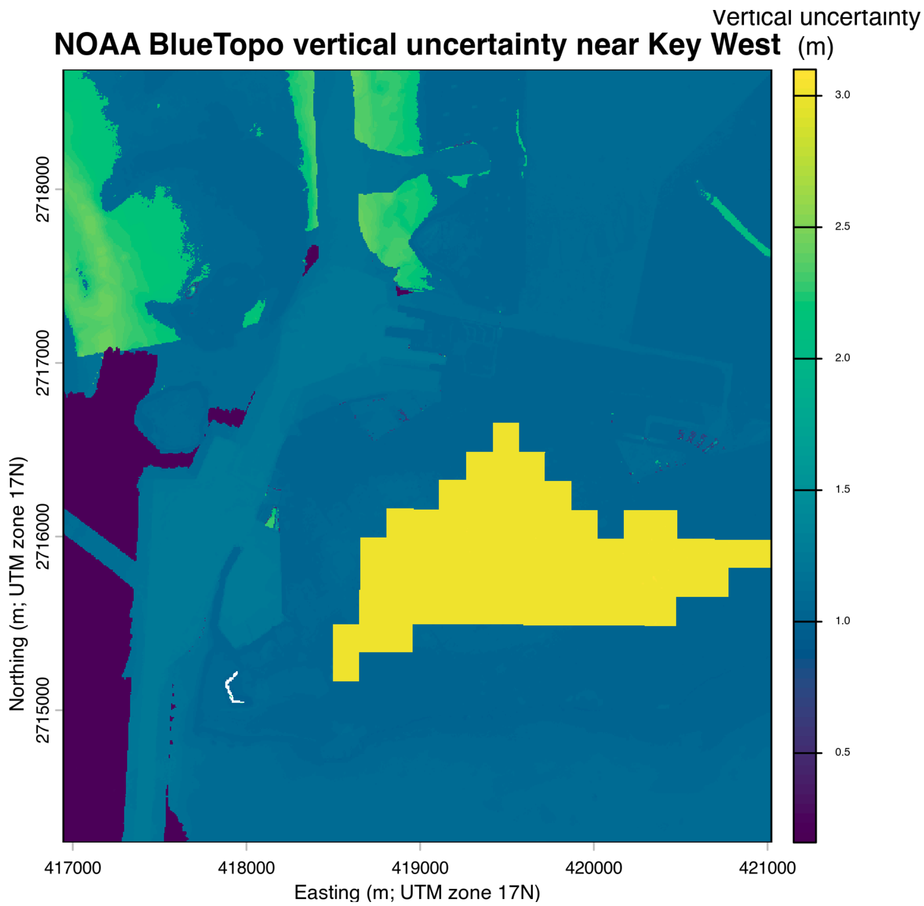

Figure 3: Associated BlueTopo vertical uncertainty for the Key West AOI in UTM zone 17N coordinates. Values are shown in metres from the source layer; this figure demonstrates that `bluertopo` can retain and open the uncertainty band with the same selected asset set.

There is not yet evidence of realized external scholarly use, citations, releases, or community engagement. The public repository history is very recent and contains one human contributor. Accordingly, this manuscript makes no adoption claim: its current significance is prospective and depends on independent use, sustained public development, and release/archiving practices.

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